Monte Carlo Method

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I. MONOCHROMATIC BEAM

We began by studying how a monochromatic beam with energy $E = 1keV$ propagates inside an infinitely thick block of aluminium. For this, it is important to use the Exponential Attenuation Law (Beer-Lambert):

$$I = I_0 e^{-\mu x} \tag{1}$$

Where I_0 is the initial number of photons in the beam before entering the block, μ is the linear attenuation coefficient in cm^{-1}, defined as $\mu_{tot} = \mu_C + \mu_R + \mu_f$, and I is the number of photons that continue to propagate after x cm of material thickness. μ_f, μ_R and μ_C represent, respectively, the attenuation coefficients of photoelectric absorption, Rayleigh scattering and Compton scattering. In the case of aluminium, $\mu_{Al} = 3196.1283$ cm^{-1}. To study the propagation of this beam, a Monte Carlo (M-C) simulation was implemented. Using the Inverse Transform Method for **Equation 1**, x_i was randomly generated following the distribution:

$$x_i = -\frac{ln(1-\xi)}{\mu} \tag{2}$$

Where $\xi \in [0;1]$ is a random number uniformly generated using `drand48()`. By generating 10^5 points from this expression and distributing them into 200 intervals with width $\Delta = 1.91296 \cdot 10^{-5}$ cm, the histogram plot was obtained as shown in **Fig. 1**. Additionally, one of the objectives to validate this method was to calculate the thickness x for which the fraction of surviving photons is half. Analytically, from **Equation 1**, it follows that $\frac{1}{2} = e^{-\mu x} \Leftrightarrow x = \frac{ln(2)}{\mu} = 2.1687 \times 10^{-4} cm$. From **Fig. 1**, it can be stated that the Monte Carlo method was successful, since the simulation shape (histogram) closely matches the theoretical curve given by **Eq.1**. Additionally, the minimum thickness x for which the surviving photon fraction is $\frac{1}{2}$ was numerically calculated, represented by the yellow line. The result obtained was $x_{M-C} = 2.18602 \times 10^{-4}$ cm, which, compared to the theoretical value $x_{teo} = 2.16871 \times 10^{-4}$ cm, yields a relative error of $\epsilon = 0.7\%$, confirming that the Monte Carlo simulation is a solid approximation.

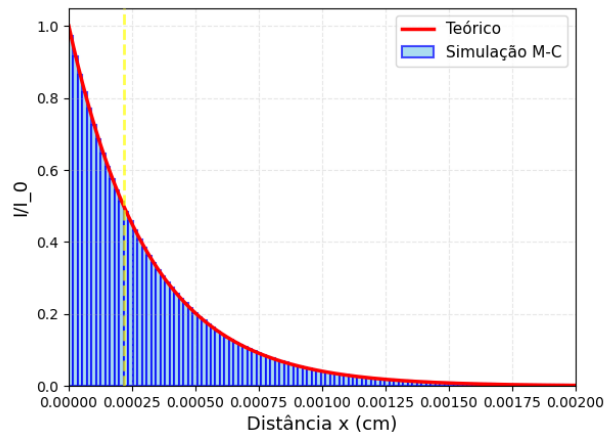

Figure 1: **Blue**: histogram of the fraction of photons from a monochromatic beam that survive absorption as a function of thickness (distance) x in the aluminium block, in cm. **Red**: theoretical curve (equation 1). **Yellow**: line indicating the x for which $\frac{I}{I_0} = \frac{1}{2}$. Results obtained for $I_0 = 10^5$, $N_{bins} = 200$ and $\Delta = ***$.

II. POLYCHROMATIC BEAM

In this section, the propagation of a polychromatic photon beam was studied, i.e., a beam with multiple energy values rather than a single one. For this type of beam, the linear attenuation coefficient cannot be obtained analytically, since its value depends on the energy of each photon. Therefore, to develop the Monte Carlo method, numerical tables obtained experimentally were used, such as `110KeV.spec`, which provides a distribution of photon fluxes N $(kV\ cm^{-2}\ mAs)$ as a function of discrete energy values between $[11;110]keV$ for a polychromatic X-ray beam passing through aluminium.

Before implementing Monte Carlo, N_i was normalised for each value of E_i, summing all values and dividing each one by that sum, thus obtaining the Probability Density Function (**PDF**). This was followed by integration over all values to obtain the Cumulative Distribution Function (**CDF**), which allows assigning a probability to each energy value and generating energies accordingly. To simulate propagation using Monte Carlo, attenuation coefficient tables were obtained from **NIST** for each energy value of a beam passing through aluminium. Since μ was given in units (ρ/cm^{-2}) and E in MeV, each μ_i was multiplied by the density $\rho_{Al} = 2.7$ gcm^{-3}, and each E_i by 10^3 to match the `110KeV.spec` table. In the Monte

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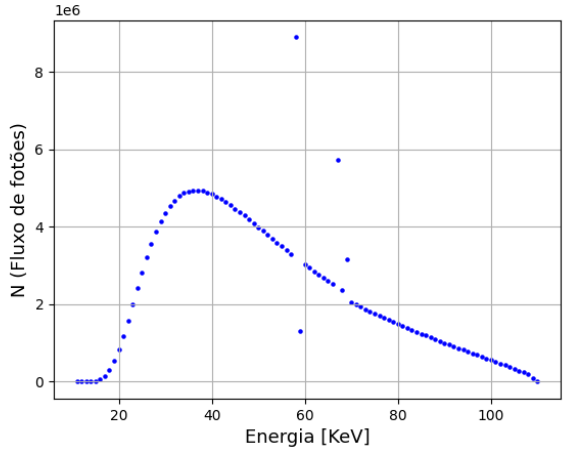

Figure 2: Plot of the data from the file `110kV.spec`, for a total flux of 250 thousand photons.

Carlo process, random numbers between 0 and 1 were generated, corresponding to the **CDF** of `110KeV.spec`, and mapped to energy values. For each generated energy, the corresponding μ was obtained from **NIST**. Finally, using expression **(2)** from the previous section, 10^5 photons were generated and distributed into 200 intervals, each with $\Delta = 0.0324\ cm$.

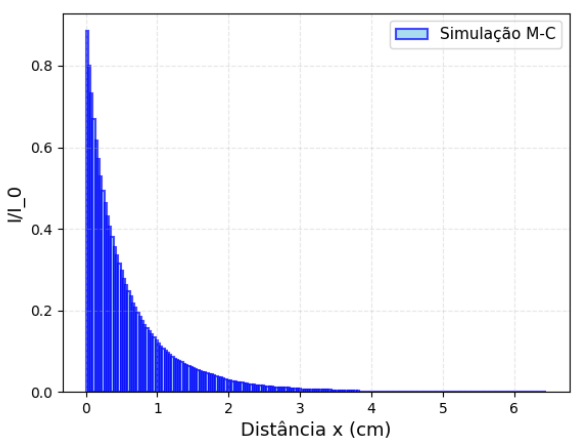

Figure 3: Histogram of the fraction of photons that survive without interaction (I/Io) as a function of x, obtained using the Monte Carlo method

Using the method from the previous section, the minimum thickness for which the photon fraction is half was obtained: $x_{2,M-C} = (0.25 \pm 0.02)\ cm$. Comparing both $x_{i,M-C}$ and the overall graph with the monochromatic case, a significant discrepancy is observed. In the monochromatic beam, all photons have the same energy, implying a constant attenuation coefficient μ throughout propagation. In contrast, in the polychromatic beam, photons with different energies coexist, each associated with its own attenuation coefficient $\mu(E)$. Higher-energy components have significantly lower μ values and therefore greater penetration capability. As the beam propagates, lower-energy components are attenuated first, leaving predominantly high-energy photons — a phenomenon known as beam hardening. This progressive enrichment in more penetrating photons results in slower overall attenuation, requiring a greater thickness to achieve the same transmission fraction as in the monochromatic case. To estimate the average total attenuation coefficient of aluminium, an *educated guess* was performed, assuming that the values in **Fig. 2** follow a distribution similar to the monochromatic beam (**Eq. 1**). Thus, a plot of $log(\frac{I}{I_0})(x)$ was performed, along with a linear fit $y = mx + b$.

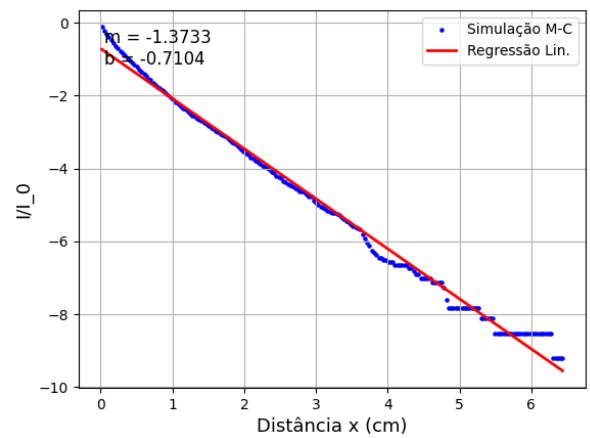

Figure 4: Plot of the linearised data and the corresponding fitted line.

The values obtained were $m = -1.373 \pm 0.02$ and $b = -0.710 \pm 0.09$, leading to $\mu_{med} = m = (1.373 \pm 0.02)\ cm^{-1}$. The parameter b accounts for systematic errors such as statistical noise in bins, approximation errors of the Monte Carlo method, fitting an effective exponential to a non-exponential behaviour, and linear regression errors. Comparing with tabulated values (**NIST**), it is again concluded that the Monte Carlo simulation was successful. Finally, the mean attenuation length is $\lambda = \frac{1}{\mu} = (0.728 \pm 0.005)\ cm$.